Industrial Internship Report on

Smart City – Forecasting of Smart city traffic patterns

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was Smart City Traffic Prediction using Machine Learning. The project focuses on predicting traffic conditions such as congestion levels and traffic flow using historical traffic data and machine learning algorithms. The system helps in forecasting traffic patterns which can assist in better traffic management and planning in smart cities. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

Smart City – Forecasting of Smart city traffic patterns

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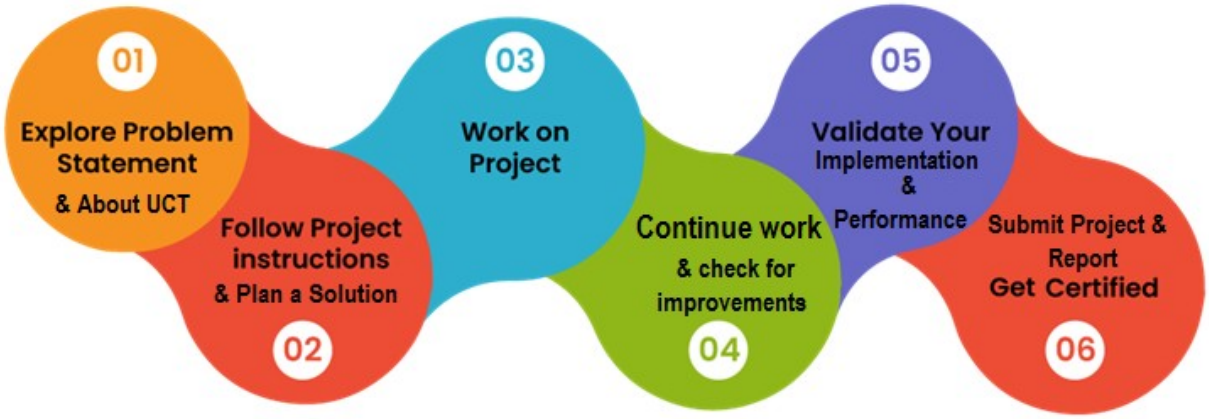
1 Preface

This report summarizes the work done during the 6 weeks of Industrial Internship program. Internships play a crucial role in career development as they provide practical exposure to real-world problems and help in enhancing technical and professional skills.

During this internship, I worked on the project “Smart City Traffic Prediction using Machine Learning,” which focuses on predicting traffic patterns and congestion using machine learning techniques.

This opportunity was provided by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd. The program was well structured with proper guidance, weekly tasks, and project development phases. During this internship, I learned data preprocessing, machine learning model building, and traffic data analysis.



I would like to thank my mentors, upskill Campus, and UniConverge Technologies Pvt Ltd for their guidance and support.

I would also like to encourage my juniors and peers to actively participate in such internships as they provide valuable practical knowledge and industry exposure.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

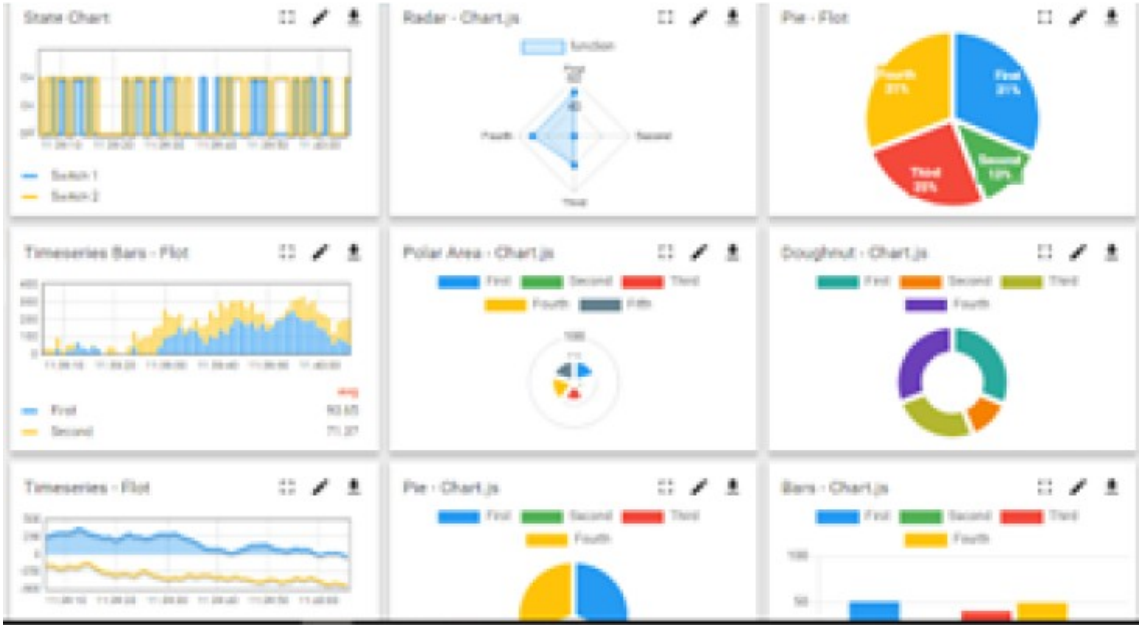
i. UCT IoT Platform (uct Insight)

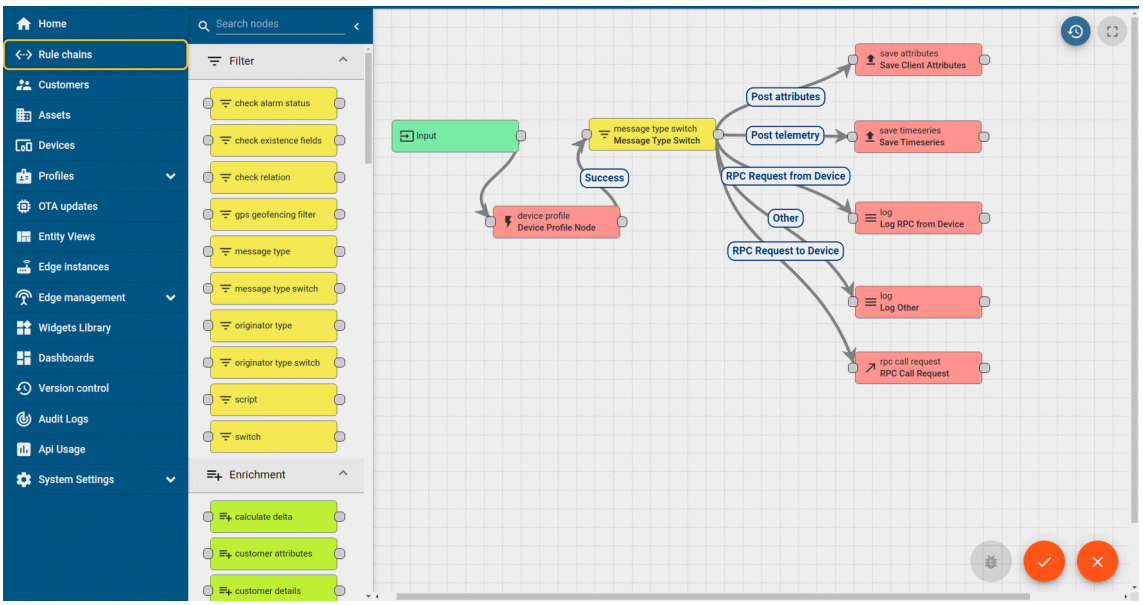
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js


Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
device profile
Device Profile Node
Success
message type switch
Message Type Switch
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii.

Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

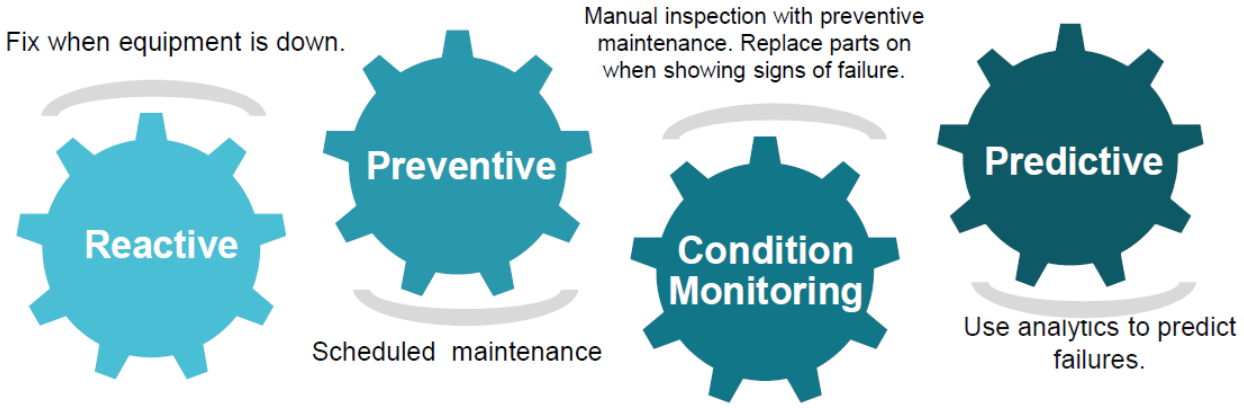
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

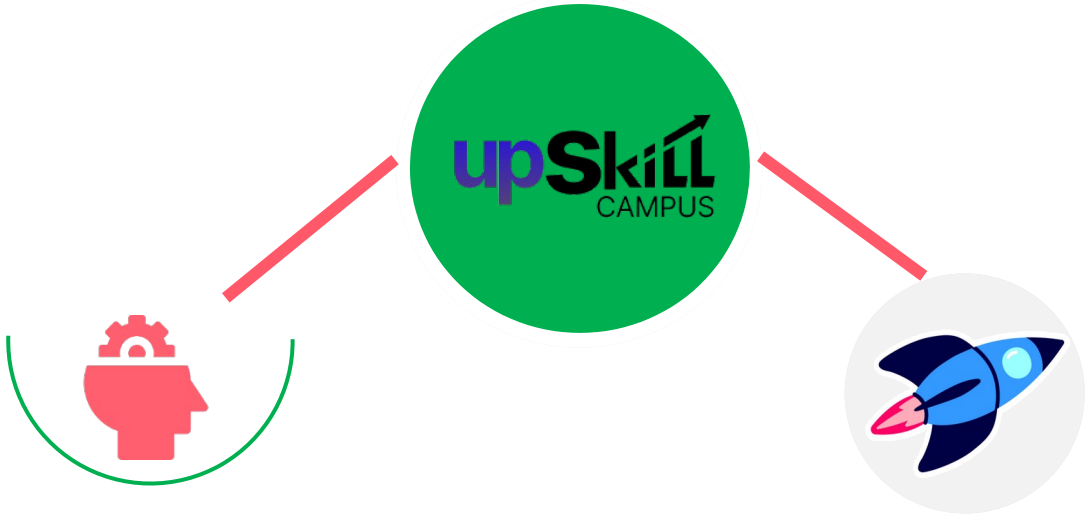
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective of this internship program was to:

- ☛ Gain practical experience in machine learning and data science
- ☛ Work on real-world traffic prediction problems
- ☛ Understand smart city traffic systems
- ☛ Improve technical skills
- ☛ Develop problem-solving abilities

2.5 Reference

[1] Python Official Documentation – https://docs.python.org/3/

[2] Pandas Library Documentation – https://pandas.pydata.org/docs/

[3] Traffic Dataset (CSV file provided during internship)

2.6 Glossary

Terms	Acronym
CSV	Comma Separated Values; a file format used to store tabular data.
EDA	Exploratory Data Analysis; the process of analyzing datasets to understand patterns and trends.
ML	Machine Learning; a technique where systems learn from data to make predictions.
Traffic Prediction	The process of forecasting traffic conditions based on historical data.

3 Problem Statement

In the assigned problem statement, the objective is to develop a Smart City Traffic Prediction System using machine learning techniques.

Traffic congestion is one of the major challenges faced by modern urban cities due to rapid urbanization and increasing number of vehicles. Inefficient traffic management leads to problems such as increased travel time, fuel wastage, environmental pollution, and road accidents.

Traditional traffic management systems mainly rely on fixed traffic signals and manual monitoring, which are not capable of handling dynamic and real-time traffic conditions. These systems lack the ability to analyze historical data and predict future traffic patterns.

To overcome these limitations, there is a need for an intelligent system that can analyze historical traffic data and predict future traffic conditions such as congestion levels and traffic flow. This prediction can help authorities in better traffic planning, signal control, and congestion management.

The proposed system uses data analysis and machine learning techniques to process traffic datasets. It identifies patterns, peak traffic hours, and trends in the data, and uses this information to predict future traffic conditions.

The system aims to support smart city development by providing data-driven insights that can improve traffic efficiency, reduce congestion, and enhance overall urban mobility.

4 Existing and Proposed solution

Existing Solutions:

In current urban traffic management systems, traffic is mainly controlled using fixed signal timings, manual monitoring, and basic surveillance systems such as CCTV cameras. Some advanced systems use sensors and IoT devices to monitor traffic flow, but many of these systems only provide real-time data without performing predictive analysis.

Limitations of Existing Solutions:

- Fixed signal systems cannot adapt to dynamic traffic conditions
- Manual monitoring is time-consuming and less efficient
- Lack of predictive capability to forecast future traffic conditions
- Inefficient handling of peak traffic hours and congestion
- Limited data analysis and poor utilization of historical data

These limitations result in increased traffic congestion, longer travel times, and inefficient traffic management.

Proposed Solution:

To overcome these limitations, the proposed system “Smart City Traffic Prediction using Machine Learning” is developed. The system uses historical traffic data stored in CSV format and applies data analysis and machine learning techniques to identify traffic patterns and trends.

The process includes data collection, preprocessing, analysis, and prediction of traffic conditions such as congestion levels and traffic flow. The system can predict future traffic situations based on past data, helping in better traffic planning and decision-making.

Value Addition:

- Provides predictive analysis instead of only real-time monitoring
- Helps in identifying peak traffic hours and congestion zones
- Improves traffic management efficiency
- Supports data-driven decision making for smart cities
- Reduces travel time and improves urban mobility

4.1 Code submission (Github link):

https://github.com/sugandharajbhar24-bit/upskillCampus/blob/main/SmartTrafficPrediction.py

4.2 Report submission (Github link) :

5. Proposed Design/ Model

The proposed system “Smart City Traffic Prediction using Machine Learning” is designed to analyze historical traffic data and predict future traffic conditions using a structured and systematic approach. The system follows multiple stages starting from data collection to final prediction and visualization.

1. Data Collection:

The system uses traffic dataset stored in CSV format. This dataset contains information such as traffic volume, time, location, and other relevant parameters required for analysis.

2. Data Preprocessing:

The collected data is cleaned to remove missing or inconsistent values. Data preprocessing also includes formatting, normalization, and handling of outliers to ensure the dataset is suitable for analysis.

3. Exploratory Data Analysis (EDA):

In this stage, the dataset is analyzed using statistical methods and visualization techniques. Graphs and charts are created to identify patterns such as peak traffic hours, congestion levels, and traffic trends.

4. Feature Selection:

Important features affecting traffic conditions are selected from the dataset. This helps in improving the accuracy and efficiency of the prediction model.

5. Model Building:

Machine learning techniques are applied to train the model using historical data. The model learns patterns and relationships between different traffic parameters.

6. Traffic Prediction:

After training, the model is used to predict future traffic conditions such as congestion levels and traffic flow based on input data.

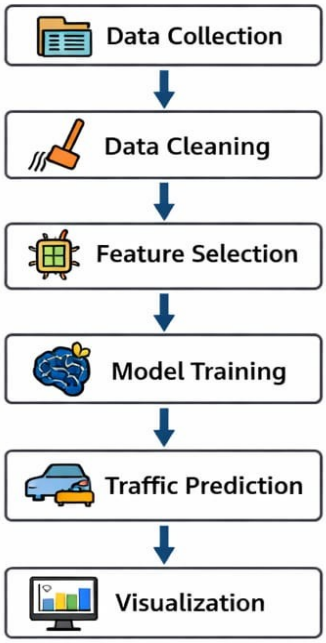
7. Visualization:

The predicted results are displayed using graphs and charts for better understanding. Visualization helps in identifying trends and making decisions easily.

Overall, the system converts raw traffic data into meaningful insights and predictions, which can be used for better traffic management in smart cities

4.3 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM



6. Performance Test

The performance testing of the system is conducted to evaluate how effectively the Smart City Traffic Prediction system works under different conditions. The main objective is to analyze the accuracy, efficiency, and reliability of the system in predicting traffic conditions using historical data.

Constraints Identified:

- Data Quality: The dataset may contain missing or inconsistent values which can affect prediction accuracy
- Processing Time: Large datasets may increase computation time
- Prediction Accuracy: The model must provide reliable and accurate results
- System Efficiency: The system should process data and generate results quickly

Handling of Constraints:

To handle data quality issues, data preprocessing techniques such as cleaning and handling missing values are applied. Efficient data processing methods are used to reduce computation time. Proper feature selection and data analysis techniques are used to improve prediction accuracy. The system is designed to handle datasets efficiently and provide quick output.

6.1 Test Plan/ Test Cases

Test Case 1: Valid Dataset Input

Input: Proper CSV file with complete data

Expected Output: Accurate traffic prediction and visualization

Test Case 2: Dataset with Missing Values

Input: CSV file containing missing entries

Expected Output: System handles missing values and still produces results

Test Case 3: Large Dataset

Input: Large volume of traffic data

Expected Output: System processes data efficiently without significant delay

Test Case 4: Invalid Input

Input: Incorrect or corrupted file

Expected Output: System shows error message and prevents processing

6.2 Test Procedure

The testing process is carried out by providing different types of datasets to the system.

First, a valid dataset is loaded to check normal system behaviour. Then datasets with missing values and large size are tested to evaluate system robustness. The system processes the data, performs analysis, and generates predictions.

The outputs are verified based on expected results to ensure correctness and efficiency.

6.3 Performance Outcome

The system performed effectively under different testing conditions. It successfully handled missing values through preprocessing techniques. The prediction results were accurate and consistent with the dataset patterns.

The system processed large datasets with acceptable performance and generated results within reasonable time. Overall, the system proved to be efficient, reliable, and suitable for traffic prediction in smart city applications.

7. My learning

During this internship, I gained valuable knowledge and practical experience in the field of data analytics and machine learning.

I learned how to work with real-world datasets, especially traffic data, and understood the importance of data preprocessing. Handling missing values, cleaning data, and preparing it for analysis helped me understand how raw data is transformed into meaningful information.

I developed skills in Exploratory Data Analysis (EDA), where I analyzed traffic patterns using graphs and charts. This helped me identify trends such as peak traffic hours and congestion levels.

I also gained knowledge of machine learning concepts, including how models are trained using historical data and how they can be used to make predictions. I understood the importance of feature selection and how it improves model performance.

In addition, I improved my programming skills in Python and learned how to use libraries such as Pandas, NumPy, and Matplotlib for data processing and visualization.

This internship also enhanced my problem-solving abilities and analytical thinking. I learned how to approach a real-world problem step-by-step, from understanding the problem statement to designing and implementing a solution.

This experience helped me understand how data-driven systems are used in smart city applications and how machine learning can be applied to solve practical problems.

8. Future work scope

The current system can be further improved and extended in several ways to enhance its functionality and effectiveness.

One major improvement can be the use of real-time traffic data instead of only historical datasets. This will allow the system to provide more accurate and dynamic predictions.

Advanced machine learning and deep learning algorithms can be implemented to improve prediction accuracy and handle more complex traffic patterns.

The system can also be integrated with smart traffic control systems such as automated traffic signals, which can adjust timings based on predicted traffic conditions.

Another enhancement can be the development of a web-based or mobile application, allowing users and authorities to easily access traffic predictions and insights.

The system can be expanded to include additional parameters such as weather conditions, road incidents, and special events, which also affect traffic patterns.

Cloud deployment can be implemented to make the system scalable and accessible from different locations.

Overall, these improvements will make the system more intelligent, efficient, and suitable for real-world smart city applications.